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Breastfeeding

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Introduction

Infant feeding started thousands of years ago , since man was created & the main type was breastfeeding. But during the 19^{th} century & the development of the new world, women went out for work & many of them neglected breastfeeding & changed to bottle feeding thier children & There was an observation that mortality & morbidity rates were higher in bottle feed children.

Exclusive breastfeeding: Both the World Health Organization (WHO) and American Academy of Pediatrics, (AAP) suggest that mothers across the globe exclusively breastfeed infants for the first six months of life. This means no other food or drink besides breast milk for the first half year of a baby's life. → with the exception of drops of syrup of vitamin

They also recommend that breastfeeding be continued for two years, with additional foods being added starting at six months.

- Breastfeeding is the natural process of providing infants with the essential nutrients.

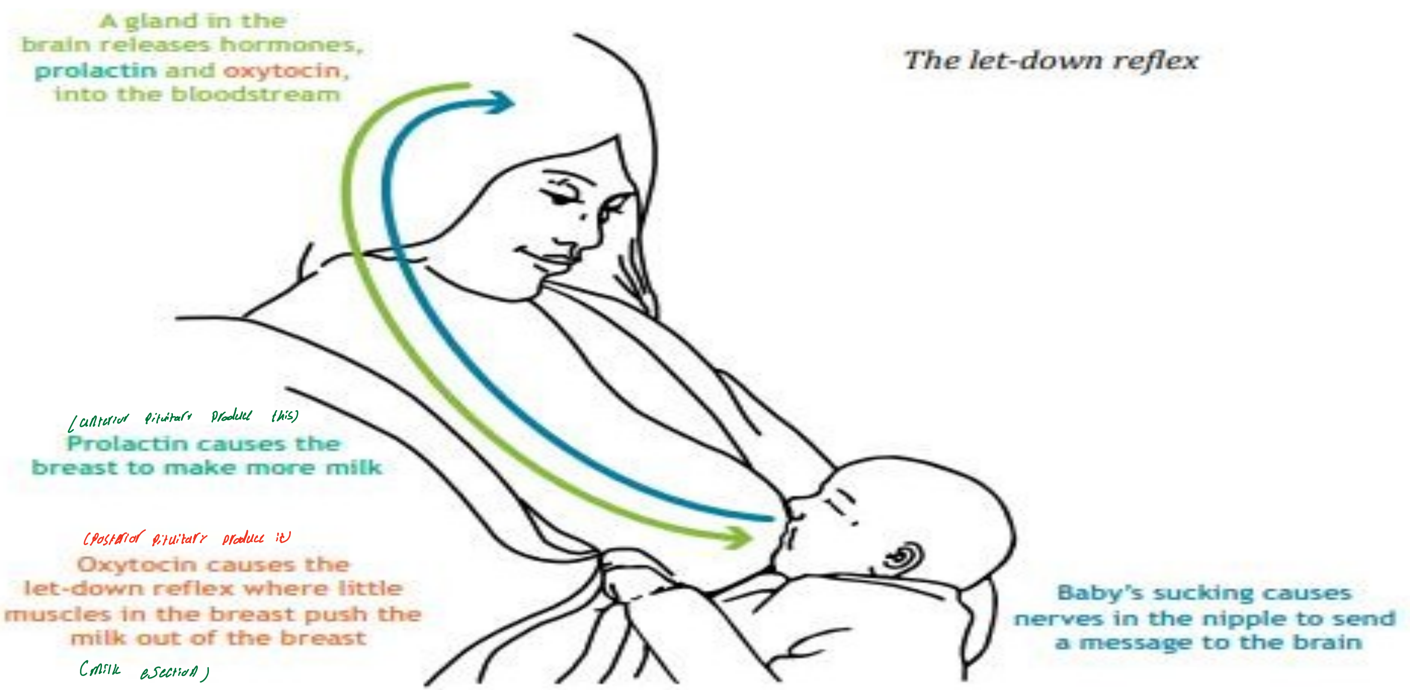
- Breastfeeding should be initiated during the first hour after birth.

- Breastfeeding is a basic right of the child as it is the best means of providing nutrition, protection, and emotional stimulation for the child’s healthy growth and development. also physical, psychological, emotional benefit to mother's, improve mother-infant bond

Global statistics (WHO 2024):

- Only about 44% of infants under 6 months are exclusively breastfed worldwide.
- WHO aims to increase exclusive breastfeeding rate to at least 70% by 2030.
- The data shows that infant mortality rates   higher in developing countries among children who have not been breastfed.

How Milk is Produced?



Factors that Stimulate the Oxycontin Reflex

1. Suckling by the baby
2. Hearing or seeing the baby
3. Thinking lovingly about the baby
4. Confidence and relaxation of the mother
5. Positive emotional state

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Factors that Inhibit the Oxytocin Reflex

1. Stress, anxiety, or fear
2. Pain (e.g., cracked nipples, difficult labor), C-section
3. Fatigue or exhaustion
4. Embarrassment or lack of privacy while breastfeeding
5. Smoking or alcohol use

- In the first few days after delivery (2-6 days) , the mother feels that her breasts are empty but in fact it is not , this is because her breasts contain a small but adequate amount of colostrum.
- Following this after about 6 days , the mother will feel a fullness in the breasts which means that milk production has started ,

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- This process can be accelerated by putting the infant on the breasts more frequently than she think , soon after birth ,(within 30 min) & to continue feeding on demand . → whenever baby show sing of hunger not necessarily crying
  - → sucking finger
  - → sucking lips
  - → mouth wandering (دەمی گەڕێتی)
- Baby's nursing stimulates prolactin – the hormone responsible for producing milk, the more baby nurses, the more prolactin will be produce, and the more milk will be ejected.

The composition of breast milk :

Breast milk composition changes to fit the baby's needs, and it can change from feed to feed and over longer periods of time as well. Breast milk is generally made up of:

87% water → since lots of water, infant don't need extra-water

7% lactose

4% fat

1% protein

The composition changes due to the need of the child and & certain factors, including the following:

1. **Stages of lactation:** 1st week colostrum, 2nd week transitional milk
   After that mature milk
2. **Time of the day:** By taking a sample at different times of the day there will be a different in the composition . Fat content is highest at mid-day.

in morning water content higher

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3. **Maternal nutrition**

since mother drink more water

4. **Climate:** In hot climate….. there will be more water content so the mother does not have to give extra water because the breast supplies this need.

since mother drink less water

In cold climates …..there will be an increase in fat content which will lead to higher energy production.

beginning later

5. **Sampling time during a given feed (fore milk & hindmilk)**
6. **Individual variation from one mother to another**

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Phases of lactation

1. Colostrum :

- first week
- Yellowish thick fluid
- Small quantity
- Higher protein , minerals & fat soluble vitamins
- Facilitate establishment of normal flora in the GIT
- Facilitate passage of meconium

↳ dark green feces that will pass excessive haemoglobin preventing physiological jaundice

- Proteins in colostrum are mainly immunoglobulins which, provide passive immunity to newborns, protecting them from infections and diseases, so for that it is called 1st vaccine.
- Contain growth factors that promote the development of various tissues and organs

2. Transitional milk:

- (7-14) days.
- Bluish thin fluid.
- Larger quantities.
- Higher also in fat & lactose.
- Less protein, more water soluble vitamins.

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3. Mature Milk

- After the 2nd week.
- Bluish thin fluid.
- Large quantities. → more nursing (more breast fed) more quantity
- Mean energy value about65-75Kcal/100ml.

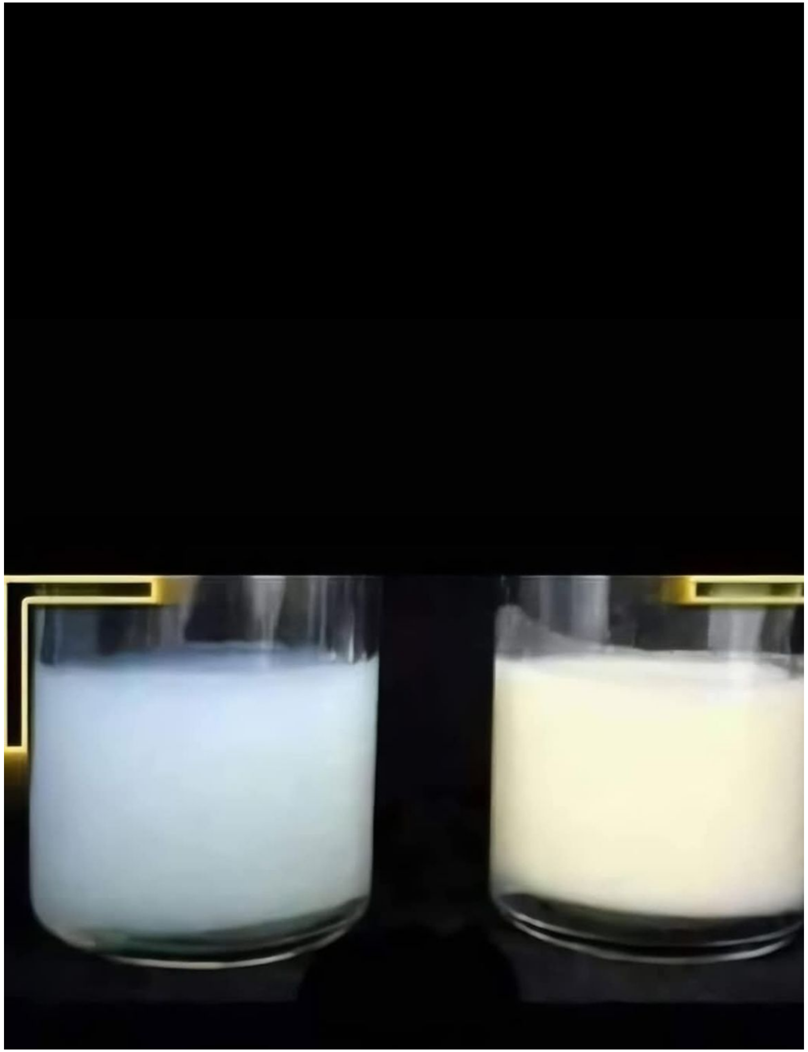
There are two types of mature milk:

- Fore-milk: This type of milk is found during the beginning of the feeding and contains water, vitamins, and protein. easy flow

first 5-minute

- Hind-milk: This type of milk occurs after the initial release of milk. It contains higher levels of fat and is necessary for growth and weight gain. slow flow difficult

after 5-minute



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since in simple word beginning of breast feeding is rapid since has no fat no nutrient but as progresses it becomes slower rate due to fat and nutrient but mother may feel the infant is fed.

to prevent this from causing fat deficiency, mother should continue feeding through one breast per feeding not alternating or stopping when hind milk begins

Breast Feeding Process

A. Preparation: The baby is placed close to the mother, with the mouth facing the nipple.

B. Position in breastfeeding

Correct positioning will determine the success of breast feeding.

1. Positioning of the mother : During breast feeding she must be in comfortable & relaxed position because if not she will be tired & end the feed quickly.

Breast Feeding Process

2. Positioning of the infant :

- The baby's neck, back and shoulders should be supported with the hand and wrist.
- baby's back lie along the arm, the baby should be facing the mother, and nose near the nipple.
- The mother should support the breast with other hand.

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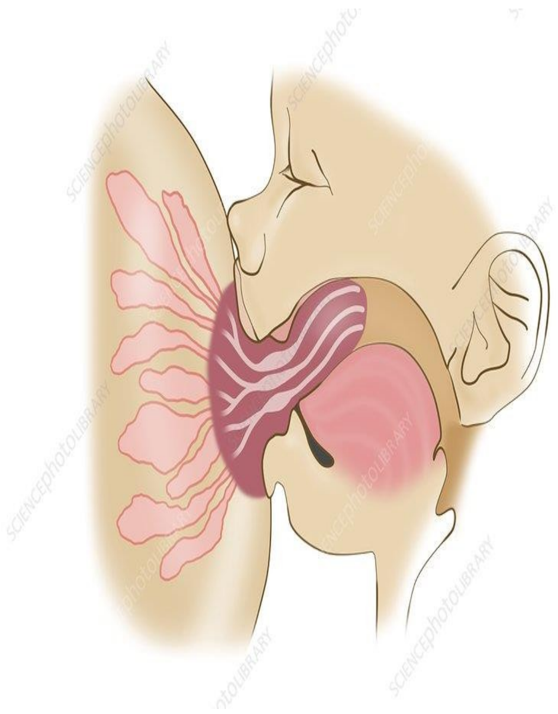
Breast Feeding Process

3. Attachment to the breast :

- His mouth should be wide open & chin touching to the breast.
- Lower lip turned outward.
- The lower part of the areola should not be seen &the upper part of the areola should be partly visible.



Breast Feeding Process

C. Reflexes:

1. Rooting reflex

The infant can find the nipple , if his cheeks or lips are stimulated (touched) , so he will open his mouth & turn his face to the nipple.

2. Sucking reflex

The infant will suckle anything that enters his mouth & touch the roof of the mouth (palate).

3. Swallowing reflex

The infant will swallow the milk present in his mouth in coordination with sucking.

Benefits of Breastfeeding:

1. Benefits for the baby:

- **Optimal Nutrition:** Breast milk provides the perfect balance of nutrients for a baby’s growth and development.
- **Immune System Support:** Breast milk contains secretory IgA antibodies that help protect the baby from infections and illnesses.
- **Digestive Health:** Breast milk is easily digestible, reducing the chances of digestive issues in infants. /easily absorbed and utilized
- **Gut Health:** The unique composition of breastmilk promotes the growth of beneficial gut bacteria, strengthening the baby's immune system.

- **Cognitive Development:** Some studies suggest that breastfeeding is associated with improved cognitive development in children.

- **Sense of Security:** Breastfeeding provides a sense of comfort and security, helping to soothe and calm the baby.

- **Reduced Risk of Allergies and Asthma:** Breastfed babies may have a lower likelihood of developing allergies and asthma.

2. Benefits for the Mother:

- Postpartum Recovery: It can help the uterus contract and reduce postpartum bleeding, aiding the mother’s recovery.
- Weight Loss: Breastfeeding can help some mothers lose pregnancy weight as it burns calories.
- Bonding and Emotional Connection: Breastfeeding fosters a strong emotional bond between the mother and baby.

- Reduced Risk of Certain Diseases: Breastfeeding is associated with a decreased risk of breast and ovarian cancer, as well as type 2 diabetes.
- Delays return to ~~ministration~~ menstruation and fertility in the mother and hence acts as a natural contraceptive. could be upto 2 years
- Breastfeeding decreased risk of postpartum depression

ABCs of breastfeeding

1. Awareness. The mother should watch for baby's signs of hunger, and breastfeed whenever the baby is hungry. This is called "on-demand" feeding.

2. Be patient. Breastfeed as long as the baby wants to nurse each time.
3. Comfort. Relax while breastfeeding, the milk is more likely to "let down" and flow.

Breastfeeding Support Systems

1. Medical Professionals: Lactation consultants, nurses, and pediatricians
   ideally in every PHC
   can provide valuable guidance and support throughout the breastfeeding journey.

2. Peer Support: Breastfeeding support groups, and mother-to-mother mentorship can offer emotional and practical support.

3. Workplace Policies: Accommodations such as lactation rooms and paid
   and day care for the infants obviously
   break times can help working mothers continue breastfeeding.